UNIT 1

DC & AC Circuits

Topics: Circuit elements (R, L, C) • Ohm's law • KCL & KVL • Series/parallel circuits • Superposition theorem • AC fundamentals • RMS, average, form & peak factor • Phasors in R, L, C • Impedance • Active, reactive & apparent power • Power factor

START HERE
Big picture: Electricity is the flow of charge. Every electrical circuit has a **source** (battery/supply), a **path** (wires) and a **load** (bulb, motor, heater). Unit I teaches the rules that tell us how much voltage and current each part gets.

1 DC CIRCUITS

1.1 Circuit Elements: R, L and C

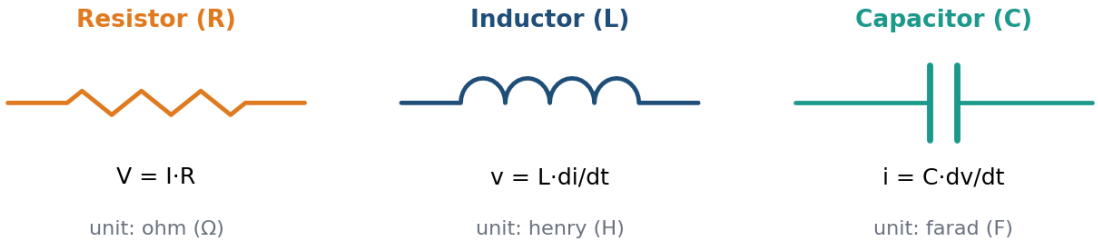


Fig 1.1 Symbols of the three passive elements

Element	What it does	Energy	Behaviour in steady DC
R Resistor	Opposes current; turns electrical energy into heat	Dissipated: $P = I^2R$	Acts as resistance R
L Inductor	Opposes *change* of current (magnetic field)	Stored: $\frac{1}{2}LI^2$ (joule)	Short circuit (plain wire)
C Capacitor	Opposes *change* of voltage (electric field)	Stored: $\frac{1}{2}CV^2$ (joule)	Open circuit (blocks DC)

DEFINITION
Resistance of a wire: $R = \rho l/A$, where ρ = resistivity (Ω·m), l = length, A = cross-section area. Longer/thinner wire → more resistance.

APPLICATIONS
R: heaters, bulbs, fuses, voltage dividers. **L:** chokes, transformers, motors, relays. **C:** filters, power-supply smoothing, fan/motor starting, tuning circuits.

1.2 Ohm’s Law and its Limitations

DEFINITION

Ohm’s Law: At constant temperature, the current through a conductor is directly proportional to the voltage across it.

$$V = I \times R \Rightarrow I = V / R \Rightarrow R = V / I$$

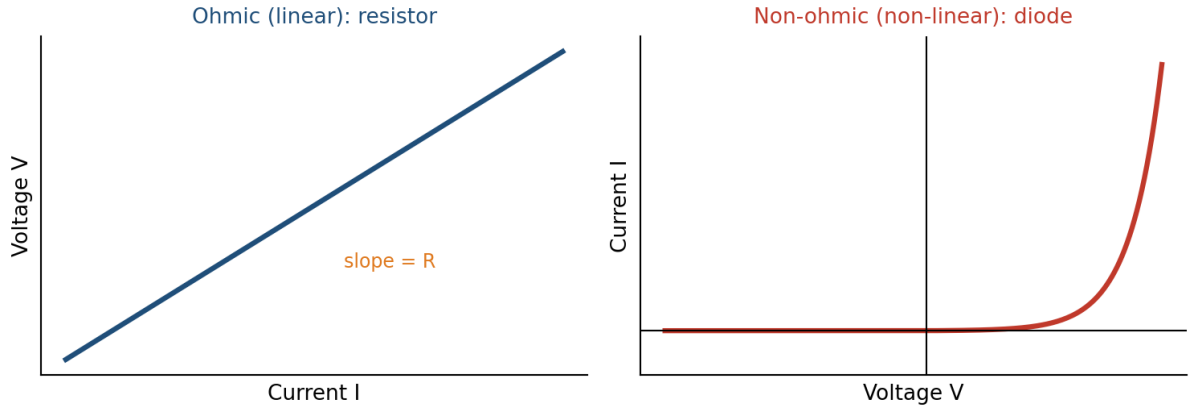


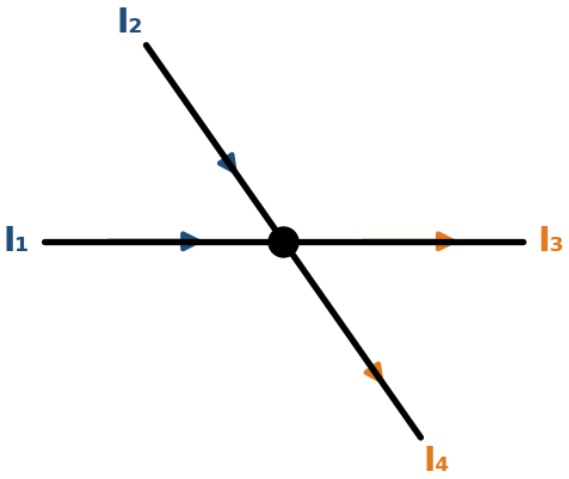
Fig 1.2 Ohmic (straight line) vs non-ohmic (curved) behaviour

Limitations of Ohm’s law

- Not valid if the **temperature** changes (resistance of metals rises with heat).
- Not applicable to **non-linear devices**: diodes, transistors, thermistors, SCRs.
- Not applicable to electrolytes, arc lamps, vacuum tubes and other **unilateral** elements.

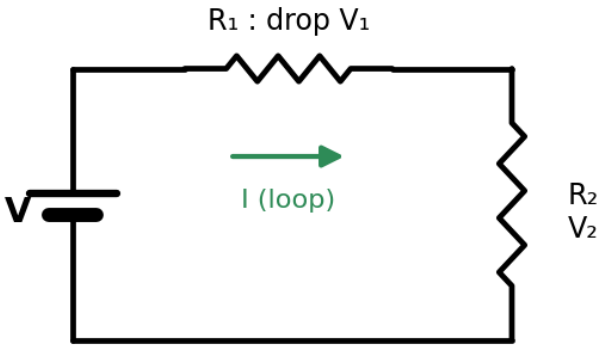
1.3 Kirchhoff’s Laws

Kirchhoff’s Current Law



KCL: $I_1 + I_2 = I_3 + I_4$
(current in = current out)

Kirchhoff’s Voltage Law



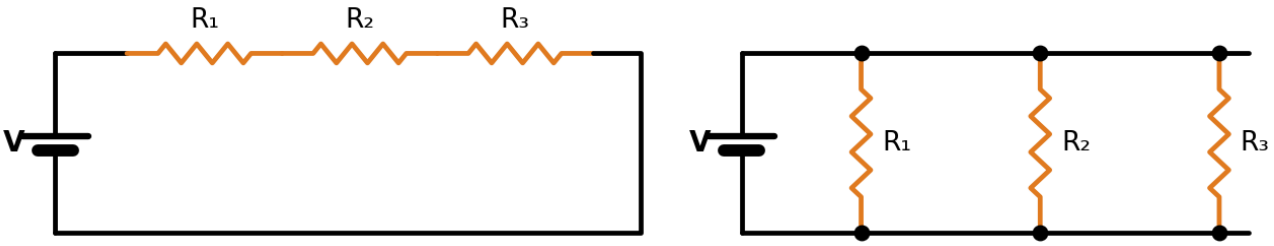
KVL: $V - V_1 - V_2 = 0$
(sum of voltages around a loop = 0)

Fig 1.3 KCL at a junction (left) and KVL around a loop (right)

	KCL (Current Law)	KVL (Voltage Law)
Statement	Algebraic sum of currents at a node = 0 (current in = current out)	Algebraic sum of voltages (rises and drops) in a closed loop = 0
Based on	Conservation of charge	Conservation of energy
Equation	$\Sigma I_{in} = \Sigma I_{out}$	$\Sigma V = 0$ (Source V = sum of drops)

EXAM TIP
Steps for KVL: (1) mark current direction in each branch, (2) choose a loop direction, (3) write drop = +IR if you go *along* the current, (4) a source is + when you pass from − to +, (5) set the sum = 0 and solve.

1.4 Series, Parallel and Series-Parallel Circuits



SERIES: same current I

R_eq = R₁ + R₂ + R₃

PARALLEL: same voltage V

1/R_eq = 1/R₁ + 1/R₂ + 1/R₃

Fig 1.4 Series and parallel connections of resistors

Property	Series	Parallel
Current	Same in all (I)	Divides: I = I₁ + I₂ + I₃
Voltage	Divides: V = V₁ + V₂ + V₃	Same across all (V)
Equivalent R	R₁ + R₂ + R₃ (larger than the biggest)	1/R₁ + 1/R₂ + 1/R₃ (smaller than the smallest)
Two resistors	—	R_{eq} = R₁R₂ / (R₁ + R₂)
Use	Fuse, switch, string lights	Household wiring (each device gets full V)

SOLVED EXAMPLE
Q1. 12 Ω, 6 Ω and 4 Ω are in parallel across 12 V. Find R_{eq} and branch currents.
$1/R_{eq}$ = 1/12 + 1/6 + 1/4 = 6/12 = 0.5 ⇒ **R_{eq} = 2 Ω**; total I = 12/2 = **6 A**. Branches: 12/12 = **1 A**, 12/6 = **2 A**, 12/4 = **3 A** (sum = 6 A ✓).
Q2 (series-parallel). 4 Ω in series with (6 Ω ∥ 12 Ω) across 24 V. 6∥12 = 4 Ω ⇒ R_{total} = 8 Ω ⇒ I = **3 A**. Voltage across the parallel part = 3 × 4 = 12 V ⇒ currents 2 A and 1 A.

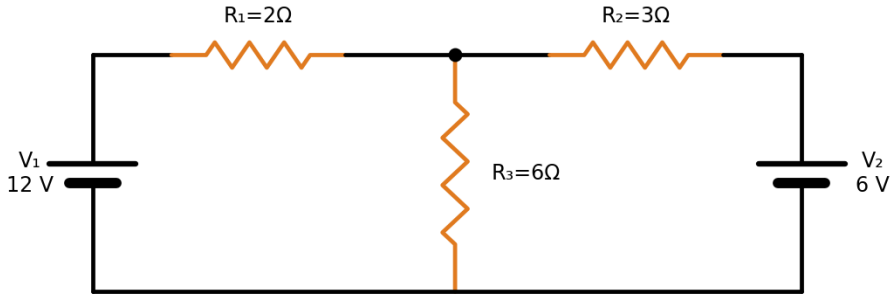
HANDY SHORTCUTS
Voltage divider (series): V₁ = V × R₁/(R₁+R₂) | **Current divider** (two parallel branches): I₁ = I × R₂/(R₁+R₂)

REMEMBER
In a **series** circuit the *largest* resistor gets the *largest* voltage; in a **parallel** circuit the *smallest* resistor carries the *largest* current. Household appliances are connected in **parallel** so that each gets full supply voltage and works independently.

1.5 Superposition Theorem

DEFINITION

In a **linear** circuit with more than one source, the current in any branch equals the **algebraic sum** of the currents produced by each source acting **alone**, with the other sources replaced by their internal resistance (**voltage source → short circuit; current source → open circuit**).



Find the current through R₃

Fig 1.5 Example circuit for superposition

SOLVED EXAMPLE

Step 1 (V_1 = 12 V only, V_2 shorted): $R_2 \| R_3 = 3\times6/9 = 2\ \Omega$; total = 2 + 2 = 4 Ω; I = 3 A; voltage across $R_3 = 3 \times 2 = 6$ V ⇒ $I_3' = 6/6 =$ **1 A**.

Step 2 (V_2 = 6 V only, V_1 shorted): $R_1 \| R_3 = 2\times6/8 = 1.5\ \Omega$; total = 3 + 1.5 = 4.5 Ω; I = 1.333 A; voltage across $R_3 = 1.333 \times 1.5 = 2$ V ⇒ $I_3'' = 2/6 =$ **0.333 A**.

Step 3 (add): $I_3 = 1 + 0.333 =$ **1.33 A** (downward). Check by node analysis: $V_A = 8$ V, 8/6 = 1.33 A ✓.

REMEMBER

Superposition works only for **linear** circuits and for current/voltage — **not for power** (power ∝ I^2, which is not linear).

2 AC CIRCUITS

2.1 AC Fundamentals

Alternating current (AC) changes its magnitude and direction periodically; the usual shape is a **sine wave**. Mains supply in India: **230 V, 50 Hz**. AC is preferred because voltage can be easily stepped up/down by transformers for efficient transmission.

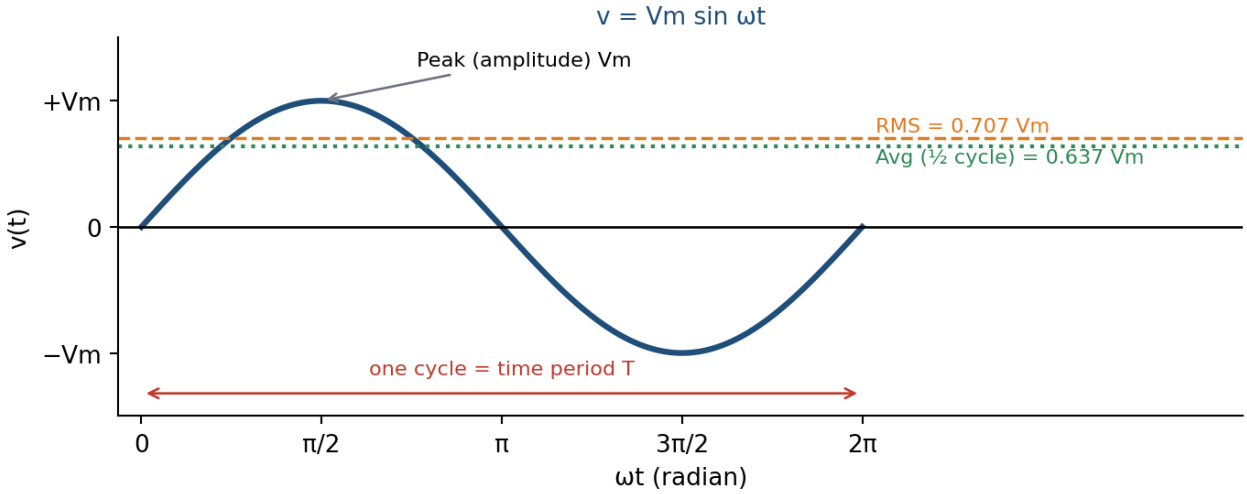


Fig 2.1 Sinusoidal waveform and its important values

$v = V_m \sin(\omega t)$ $i = I_m \sin(\omega t \pm \varphi)$ $\omega = 2\pi f = 2\pi/T$

Term	Meaning	Unit
Amplitude (peak)	Maximum value V_m or I_m	V or A
Time period T	Time for one complete cycle	s
Frequency f	Cycles per second, f = 1/T	Hz
Phase	Angular position of the wave at a given instant (ωt ± φ)	rad / degree
Phase difference φ	Angle by which one wave leads/lags another (Fig 2.2)	degree

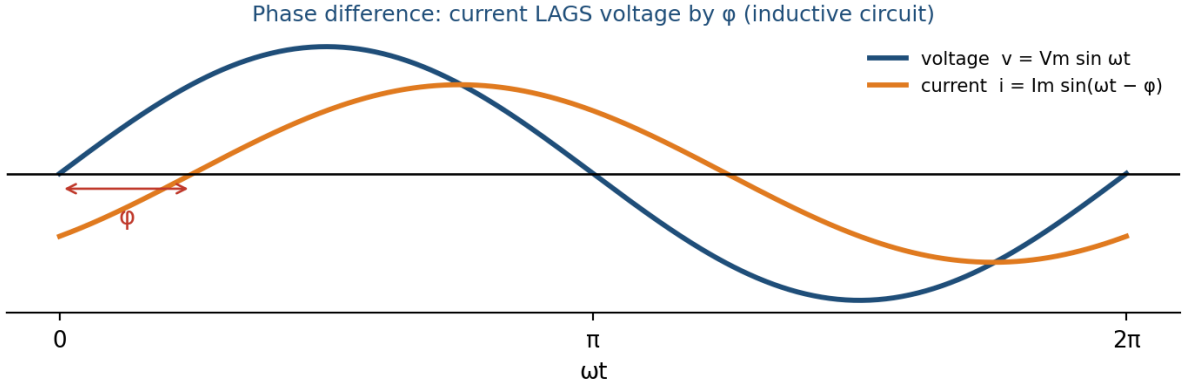


Fig 2.2 Phase difference between voltage and current

2.2 Average, RMS, Form Factor and Peak Factor

Quantity	Definition	Sine wave
Average value	Mean of instantaneous values over a *half* cycle (full-cycle average = 0)	$V_{av} = 2V_m/\pi =$ **0.637 V_m**
RMS (effective) value	Square root of mean of squares; the DC value giving the same heating effect	$V_{rms} = V_m/\sqrt{2} =$ **0.707 V_m**
Form factor	RMS value / Average value	**1.11**
Peak factor (crest)	Peak value / RMS value	**1.414**

SOLVED EXAMPLE

Q3. v = 311 sin 314t. Find V_{rms}, f and T.
$V_m = 311$ V ⇒ $V_{rms} = 0.707 \times 311 =$ **220 V**; ω = 314 ⇒ f = 314/2π = **50 Hz**; T = 1/50 = **20 ms**. Average (half-cycle) = 0.637 × 311 = 198 V.

2.3 AC through R, L and C (Phasor Diagrams)

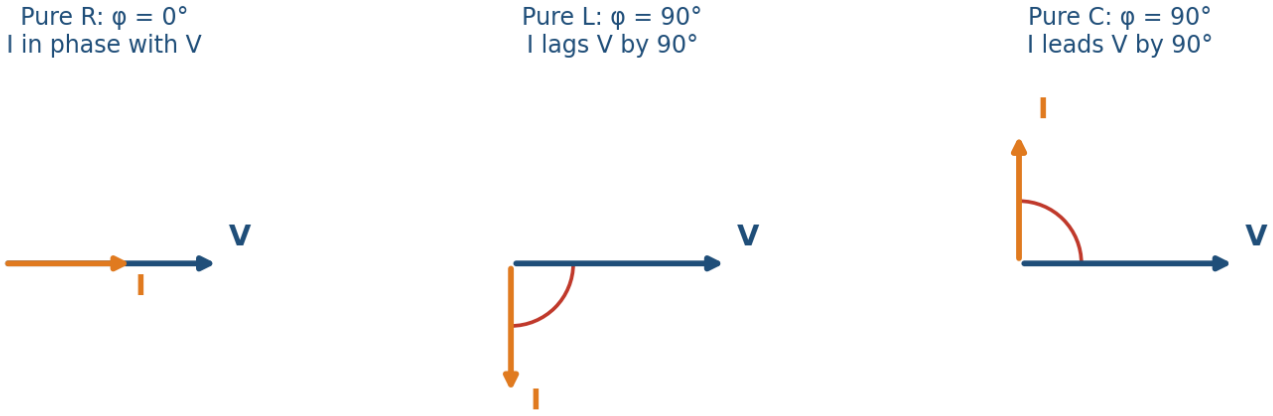


Fig 2.3 Phasor diagrams (voltage V is the reference along the x-axis)

Circuit	Opposition	Phase relation	Memory trick
Pure R	Z = R	I in phase with V (φ = 0°)	—
Pure L	$X_L = \omega L = 2\pi fL$	I **lags** V by 90°	"ELI": E before I in L
Pure C	$X_C = 1/\omega C = 1/2\pi fC$	I **leads** V by 90°	"ICE": I before E in C

Concept of Impedance (Z): the total opposition of a circuit to AC, combining resistance and reactance. For a series R-L-C circuit:

$$\mathbf{Z = \sqrt{[\ R^2 + (X_L - X_C)^2\ ]}\ \tan \varphi = (X_L - X_C) / R\ I = V / Z}$$

If $X_L > X_C$ the circuit is **inductive** (current lags); if $X_C > X_L$ it is **capacitive** (current leads); if $X_L = X_C$ it is **resonance** (Z = R, minimum).

SOLVED EXAMPLE

Q4. Find X_L of a 0.1 H coil and X_C of a 100 μF capacitor at 50 Hz.
$X_L = 2\pi \times 50 \times 0.1 =$ **31.4 Ω**; $X_C = 1/(2\pi \times 50 \times 100\times10^{-6}) =$ **31.8 Ω**. Note: X_L *increases* with frequency, X_C *decreases* with frequency (a capacitor passes high frequencies, an inductor blocks them).

APPLICATIONS

Series RLC resonance occurs at $f_0 = 1/(2\pi\sqrt{LC})$, where $X_L = X_C$, Z = R (minimum), current is maximum and pf = 1 — used in radio tuning.

2.4 Active, Reactive and Apparent Power; Power Factor

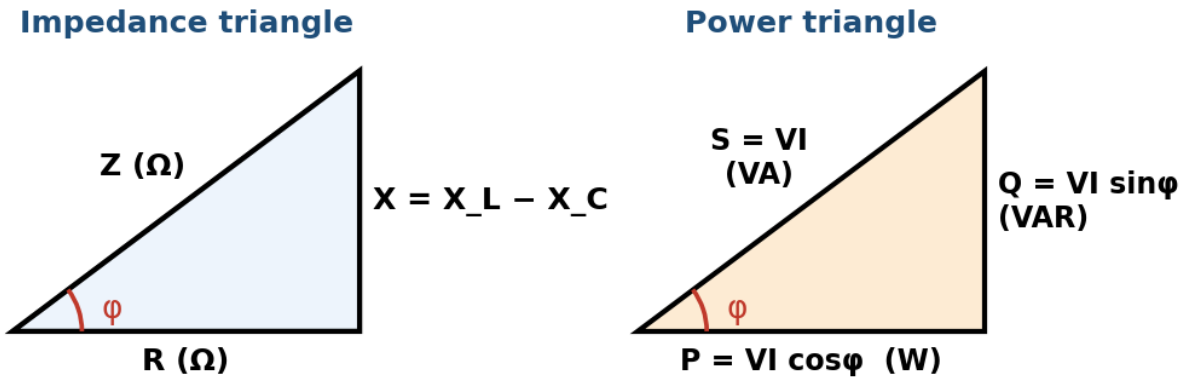


Fig 2.4 Impedance triangle and power triangle

Power	Symbol & formula	Unit	Meaning
Active (true)	P = VI cosφ = I²R	watt (W)	Real work done (heat, light, motion)
Reactive	Q = VI sinφ = I²X	VAR	Energy swapped between source and L/C; no work
Apparent	S = VI = I²Z	VA	Total power supplied; S² = P² + Q²

DEFINITION

Power factor = cos φ = R/Z = P/S (lies between 0 and 1; lagging for inductive loads, leading for capacitive loads). A **low power factor** means higher current for the same power → more losses and larger cables; it is improved by connecting **capacitors** across inductive loads.

SOLVED EXAMPLE

Q5. A series circuit has R = 30 Ω and X_L = 40 Ω across 230 V, 50 Hz. Find Z, I, pf, P, Q, S.
Z = √(30² + 40²) = **50 Ω**; I = 230/50 = **4.6 A**; φ = $\tan^{-1}(40/30)$ = 53.1°; pf = 30/50 = **0.6 lagging**.
P = 230 × 4.6 × 0.6 = **635 W** (check: I²R = 21.16 × 30 = 635 ✓); Q = 230 × 4.6 × 0.8 = **846 VAR**; S = 230 × 4.6 = **1058 VA**.

3 QUICK REVISION & EXAM CORNER

Topic	Key formulas
Ohm’s law	$V = IR$
Series / Parallel	$R_{eq} = \Sigma R$ / $1/R_{eq} = \Sigma(1/R)$
KCL / KVL	$\Sigma I = 0$ at node / $\Sigma V = 0$ in loop
AC wave	$\omega = 2\pi f$, $T = 1/f$, $V_{rms} = 0.707V_m$, $V_{av} = 0.637V_m$, FF = 1.11, PF(peak) = 1.414
Reactance & Z	$X_L = 2\pi fL$, $X_C = 1/2\pi fC$, $Z = \sqrt{(R^2 + X^2)}$
Power	$P = VI\cos\varphi$, $Q = VI\sin\varphi$, $S = VI$, $pf = \cos\varphi$

Short questions (2 marks)

- State Ohm’s law. Mention two limitations.
- State KCL and KVL.
- Define RMS value, form factor and peak factor of a sine wave.
- What is the phase relation between V and I in a pure inductor and a pure capacitor?
- Define impedance and power factor.
- Why is the RMS value used to rate AC supplies?

Long questions (5–10 marks)

- Explain superposition theorem with a neat example circuit.
- Derive RMS and average values of a sinusoidal current.
- Draw phasor diagrams for pure R, L and C circuits and explain.
- Explain active, reactive and apparent power with the power triangle.
- Three resistors 5 Ω, 10 Ω and 20 Ω are connected in (a) series and (b) parallel across 40 V. Find total current in each case. *(Ans: 1.14 A; 14 A)*

Practice numericals (with answers)

No.	Problem	Answer
1	Two resistors 6 Ω and 3 Ω are in parallel across 18 V. Find total current.	9 A (R_{eq} = 2 Ω)
2	Find RMS and average value of a sine wave with peak 200 V.	141.4 V; 127.4 V
3	R = 8 Ω in series with X_L = 6 Ω across 100 V. Find I and pf.	10 A; 0.8 lag
4	A load draws 5 A at 230 V with pf 0.8. Find P and Q.	920 W; 690 VAR
5	Find the series equivalent of 10 Ω, 15 Ω and 25 Ω, and current at 100 V.	50 Ω; 2 A

Quick MCQs

Question	Options	Ans
1. Unit of inductance is	(a) farad (b) henry (c) ohm (d) weber	b
2. Which law is based on conservation of charge?	(a) KVL (b) Ohm (c) KCL (d) Faraday	c
3. Form factor of a sine wave is	(a) 1.414 (b) 1.11 (c) 0.707 (d) 0.637	b
4. In a pure capacitor, current ... voltage by 90°	(a) lags (b) leads (c) is in phase with (d) is zero	b
5. Power factor of a pure inductor is	(a) 1 (b) 0.5 (c) 0 (d) infinity	c
6. Voltage source is replaced by ... in superposition	(a) open circuit (b) short circuit (c) resistor (d) capacitor	b